

CS & IT ENGINEERING



Operating System

Memory Management

Lecture – 7

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Recap of Previous Lecture



doubt?

Topic

Page Replacement Policies

Topic

Frame Allocation



Topics to be Covered



Topic

Multilevel Paging





Topic : Let's Take a Simple Example

$$\left. \begin{array}{l} \text{Process Size} = 32\text{B} \\ \text{Page size} = 4\text{B} \end{array} \right\} \text{no. of pages} = \frac{32\text{B}}{4\text{B}} = 8$$

Page table entry size = 1B

$$\text{Page Table Size} = ? \quad 8 * 1\text{B} = 8\text{B}$$

$$\text{no. of pages in page table} = \frac{8\text{B}}{4\text{B}} = 2$$



Topic : Now What?



Can you keep entire page table in single page?

If not then how to access the specific entry?





Topic : Page Table in Memory

Process
Page 000
Page 001
Page 010
Page 011
Page 100
Page 101
Page 110
Page 111

Page Table

000	0011
001	0111
010	1100
011	1001
100	0110
101	1111
110	1011
111	0001

Page 0 of P.T.
Page 1 of P.T.

Page table of page table

0	0100
1	1101

(outer P.T.)

Page 0 of P.T.

00	0011
01	0111
10	1100
11	1001

Page 1 of P.T.

00	0110
01	1111
10	1011
11	0001

frame no. mm
16 frames

0000	
0001	Page 111
0010	
0011	Page 000
0100	Page 0 of P.T.
0101	
0110	Page 100
0111	Page 001
1000	
1001	Page 011
1010	
1011	Page 110
1100	Page 010
1101	Page 1 of P.T.
1110	
1111	Page 101



Topic : Multilevel Paging

How to search in P.T. with page no.

CPU generates L.A.

1	0	1	
p		d	

P1		P2	d
1	0	1	

helps to
search in outer
P.T.

⇒ frame no. of
inner P.T.



on this frame
P.T. entry no. 01
is your entry.

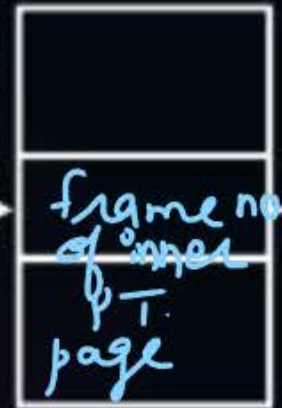


Topic : Multilevel Paging

logical address



P_1 {



outer -page table

P_2 {



outer -page table

d {



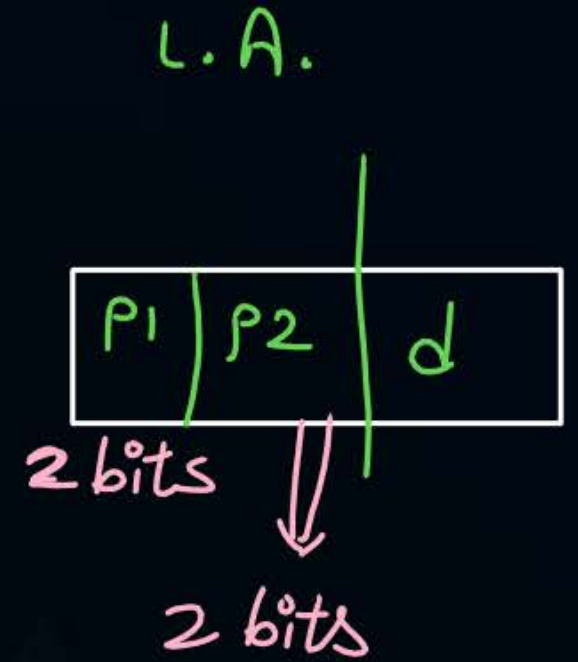
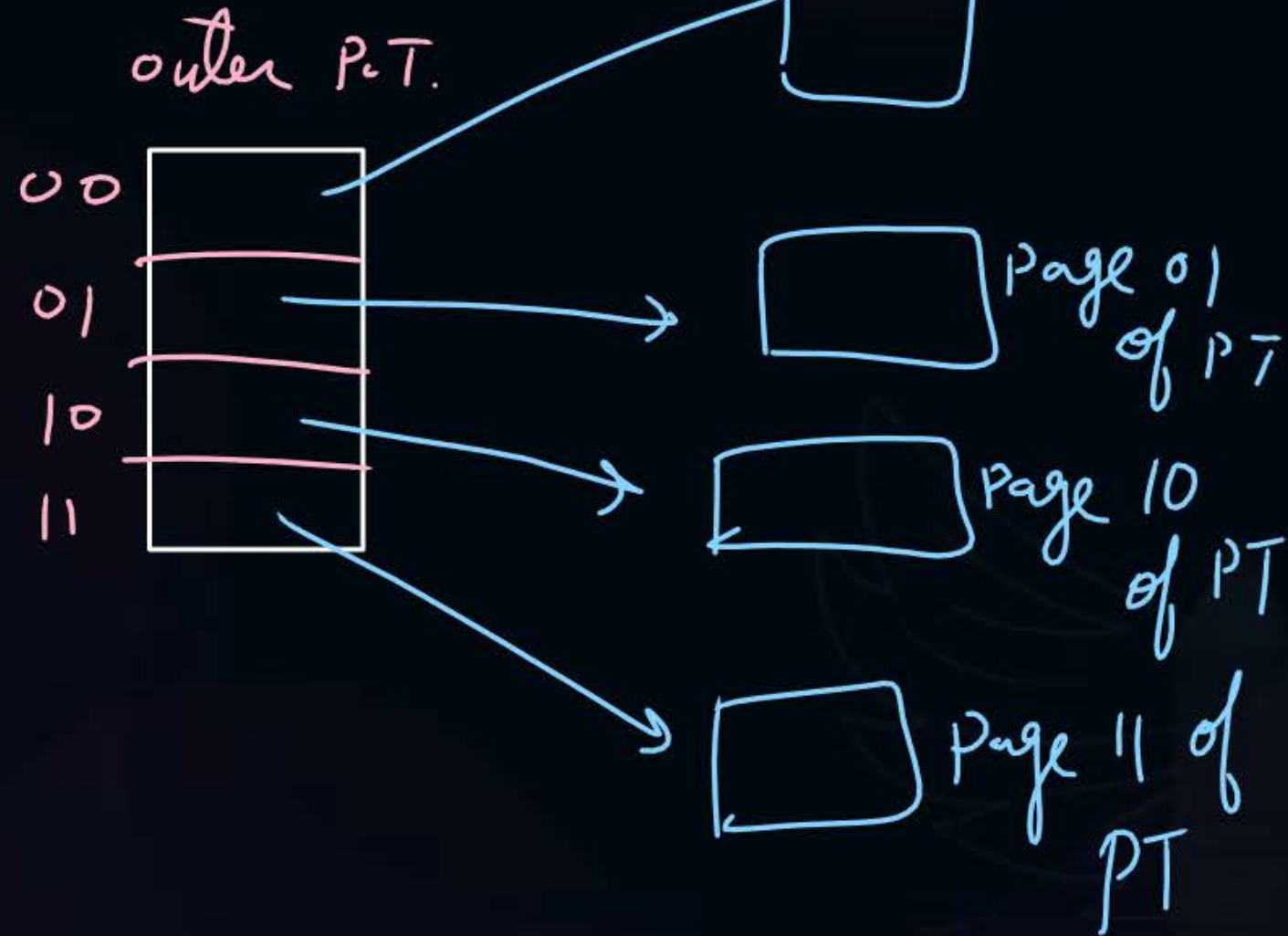
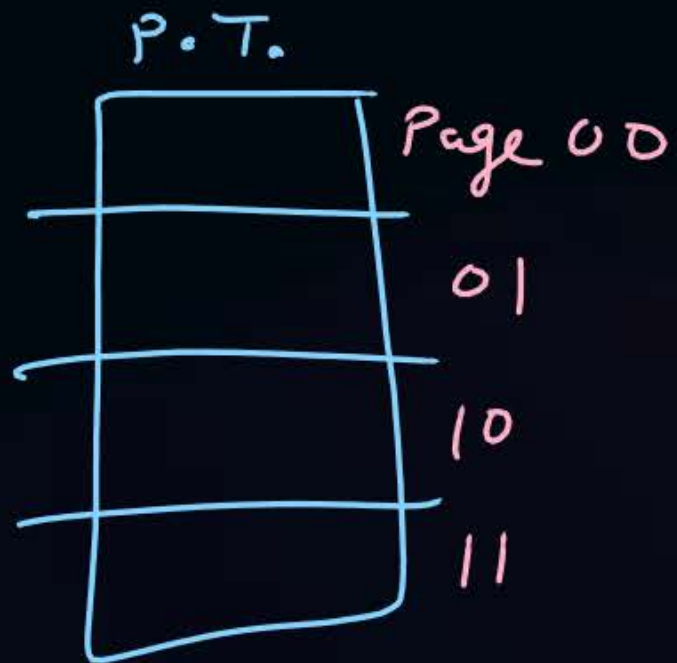
desired page





Topic : Multilevel Paging

if P.T. needs 4 pages to store



⇒ we increase P.T. levels when outer P.T. can not fit on single page.

⇒ The levels are increased till outermost P.T. does not fit in single page.



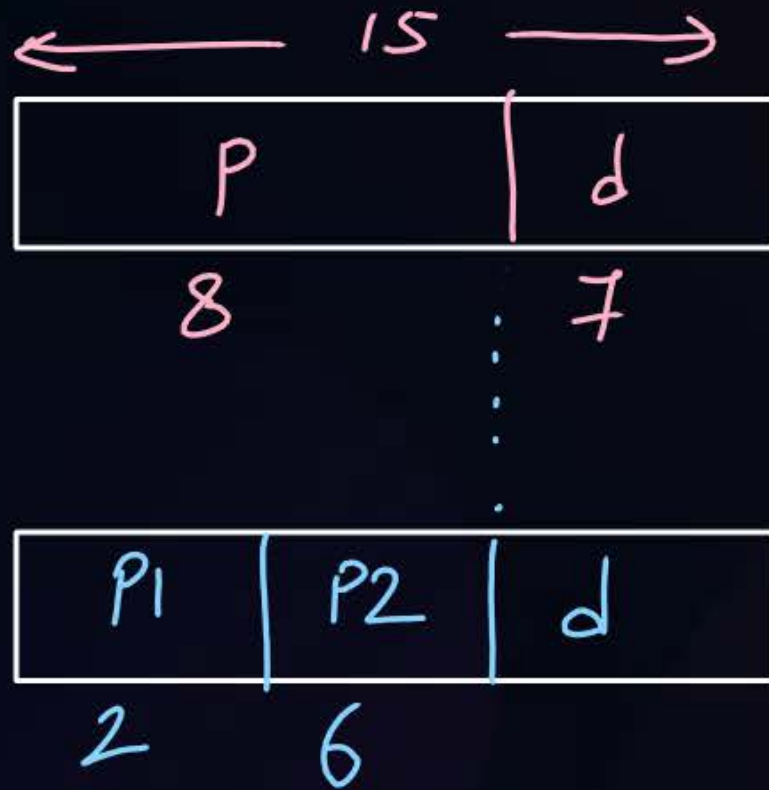
Topic : Question

VA = 15 bits

Page size = 128 bytes = $2^7 \text{ B} \Rightarrow d = 7 \text{ bits}$

Page table entry size = 2 bytes

Number of levels in multilevel page 2?



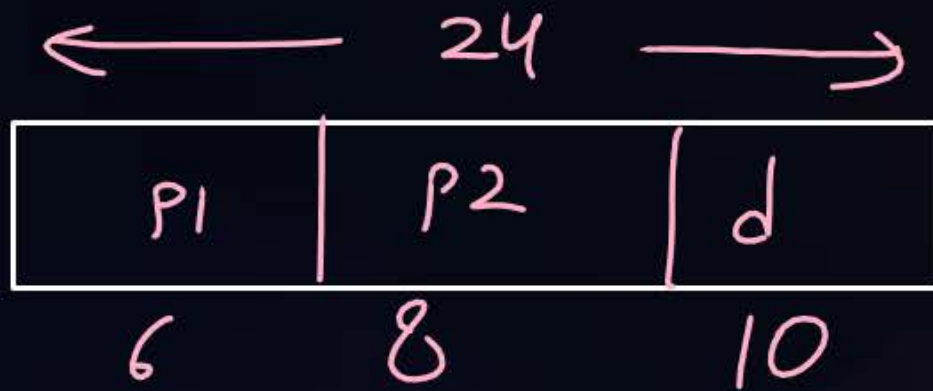
$$\begin{aligned} \text{no. of P.T. entries in one page} &= \frac{\text{Page size}}{\text{P.T.E. size}} \\ &= \frac{128 \text{ B}}{2 \text{ B}} \\ &= 64 = 2^6 \end{aligned}$$



Topic : Question

VA = 24 bits
Page size = 1Kbytes $= 2^{10} B \Rightarrow d = 10$
Page table entry size = 4 bytes

Number of levels in multilevel page 2?



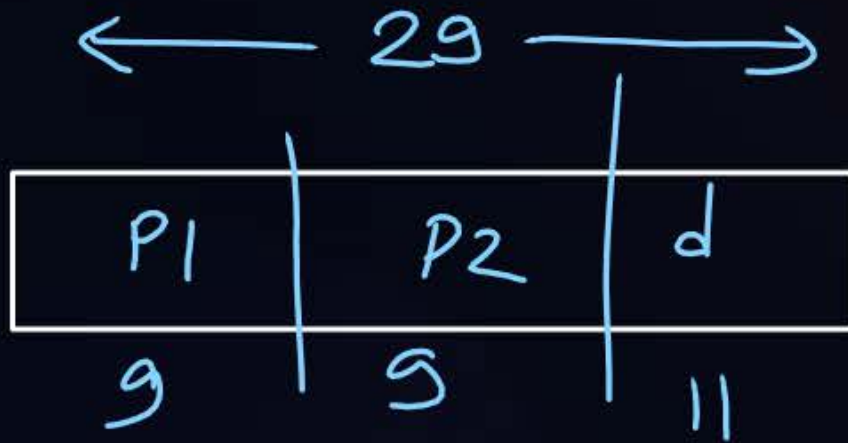
$$\begin{aligned} \text{no. of P.T. entries in one page} &= \frac{1KB}{4B} \\ &= 2^8 \end{aligned}$$



Topic : Question

VA = 29 bits
Page size = 2Kbytes = 2^{11} B $\Rightarrow d = 11$
Page table entry size = 4 bytes

Number of levels in multilevel page 2?



$$\text{no. of P.T. entries per page} = \frac{2KB}{4B} = \frac{2^{11}}{2^2} = 2^9$$



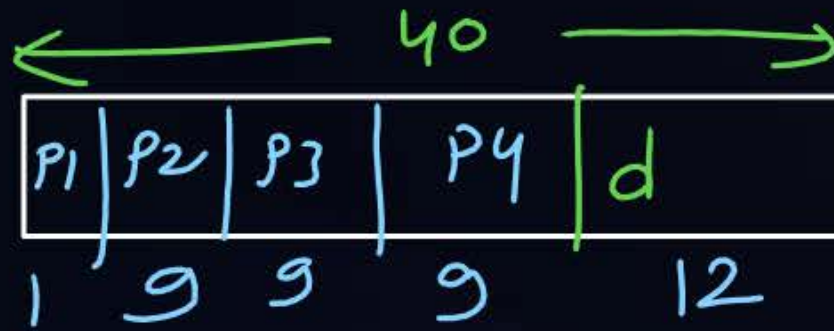
Topic : Question

VA = 40 bits

Page size = 4Kbytes = 2^{12} B $\Rightarrow d = 12$ bits

Page table entry size = 8 bytes

Number of levels in multilevel page 4?



$$\begin{aligned} \text{no. of P.T. entries per page} &= \frac{4 \text{ KB}}{8 \text{ B}} \\ &= \frac{2^{12}}{2^3} = 2^9 \end{aligned}$$

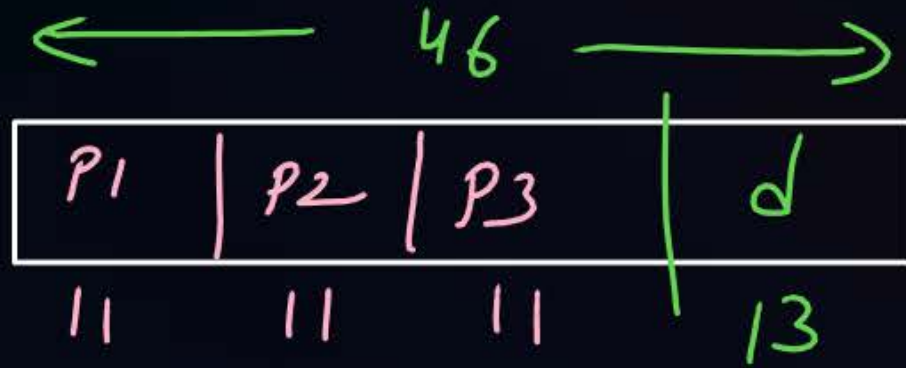


Topic : Question

Ans = 3

#Q. Consider a virtual memory system with physical memory of 8GB, a page size of 8KB and 46-bit virtual address. Assume every page table exactly fits into a single page. If page table entry size is 4B then how many levels of page tables would be required.

$d = 13$



$$\begin{aligned} \text{no. of P.T. entries per page} &= \frac{8 \text{ KB}}{4 \text{ B}} \\ &= 2 \text{ K} \\ &= 2^{11} \end{aligned}$$

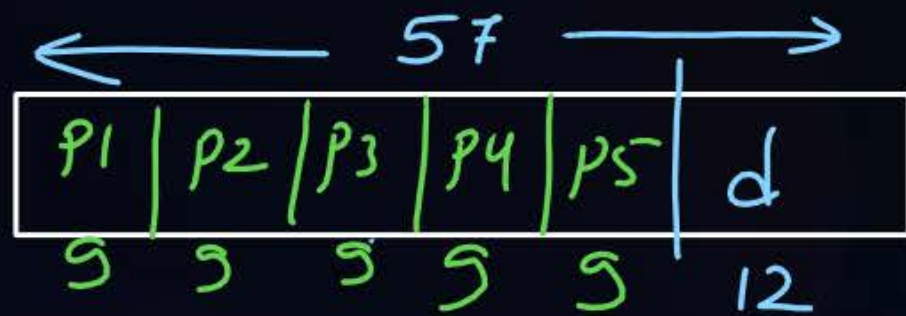


Topic : Question

GATE PYQ

- #Q. Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4KB (1KB=1024B) and a page table entry at any of the levels occupies 8 bytes. The value of L is 5?

$\rightarrow d = 12 \text{ bits}$



$$\begin{aligned} \text{no. of P.T. entries per page} &= \frac{4\text{KB}}{8\text{B}} \\ &= 2^9 \end{aligned}$$



Topic : Question

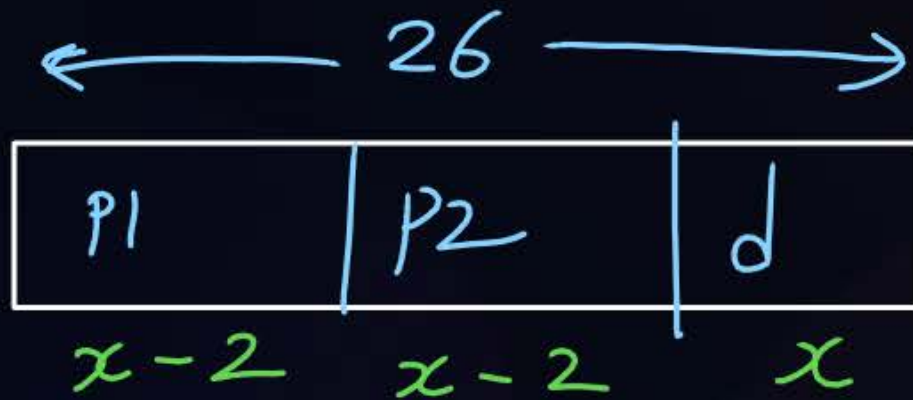
VA = 26 bits

Page table entry size = 4 bytes

2-level paging

Outer page table fits into a page exactly

Page size = 1 Kbytes?



$$x-2 + x-2 + x = 26$$

$$3x - 4 = 26$$

$$3x = 30 \Rightarrow x = 10 \Rightarrow$$

$$\text{page size} = 2^{10} \text{ B} = 1 \text{ KB}$$

Assume,

$$\text{Page size} = 2^x \text{ B} \Rightarrow d = x \text{ bits}$$

$$\begin{aligned} \text{no. of P.T. entries} &= \frac{2^x \text{ B}}{4 \text{ B}} \\ \text{per page} & \end{aligned}$$

$$\begin{aligned} &= \frac{2^x}{2^2} \\ &= 2^{x-2} \end{aligned}$$



Topic : Question

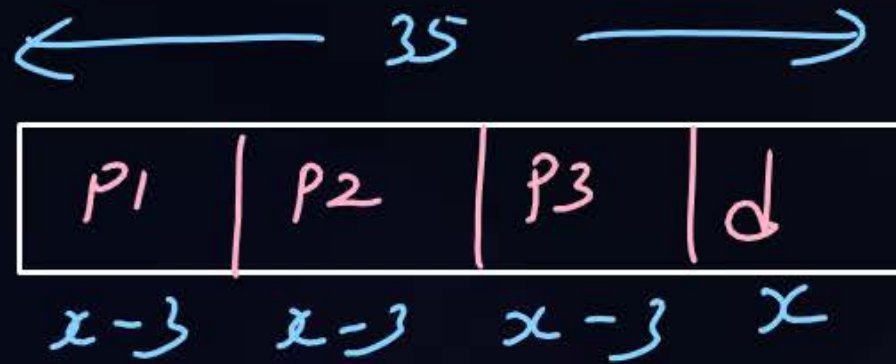
VA = 35 bits

Page table entry size = 8 bytes

3-level paging

Outer page table fits into a page exactly

Page size = 2 Kbytes?



$$3(x-3) + x = 35$$

$$4x = 44$$

$$x = 11$$

$$\text{Page size} = 2^{11} \text{ B} = 2 \text{ KB}$$

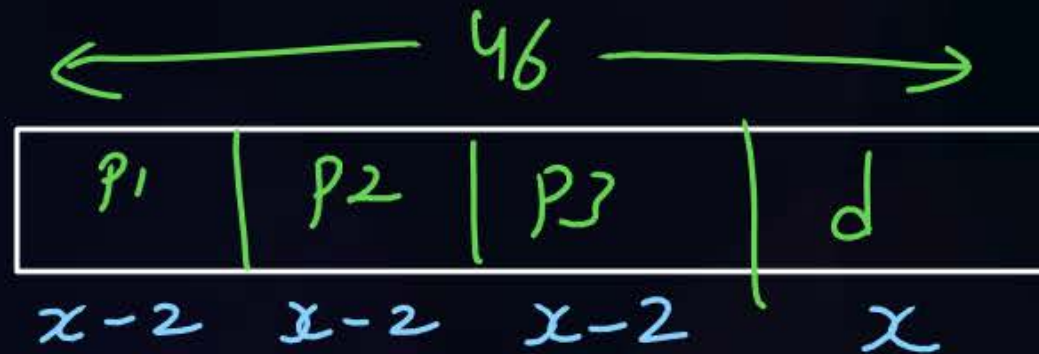


Topic : Question

Ans = 8

#Q. A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T1), which occupies exactly one page. Each entry of T1 stores the base address of a page of the second-level table (T2). Each entry of T2 stores the base address of a page of the third-level table (T3). Each entry of T3 stores a page table entry (PTE). The PTE is 32 bits in size. What is the size of a page in KB in this computer?

↓
4 Bytes



$$3(x-2) + x = 46$$

$$4x - 6 = 46$$

$$x = 13 \Rightarrow \text{Page size} = 2^{13} \text{ B} = 8 \text{ KB}$$



Topic : Question

Size of page tables across all levels?

P1	P2	D
1	8	10

Page size = 1 KB

No. of pages in outermost P.T. = 1
No. of pages in Inner P.T. = $2^{P_1} = 2^1 = 2$

Total no. of pages to store entire P.T. across all levels = 3

Total space needed to store P.T. across all levels = $3 \times 1 \text{ KB}$
= 3 KB



Topic : Question

Size of page tables across all levels?

P1	P2	D
2	9	12

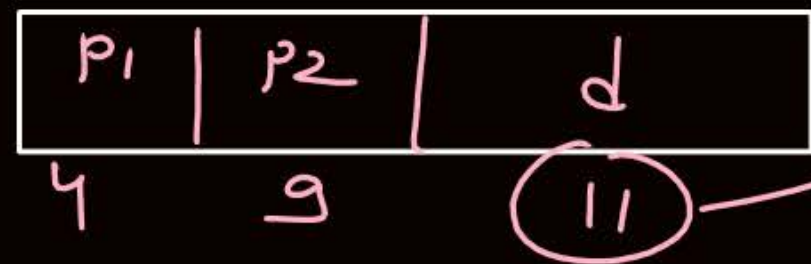
$\Rightarrow \text{Page Size} = 2^{12} \text{ B} = 4 \text{ KB}$

No. of pages to store outer P.T. = 1
|| Inner P.T. = $2^{P_1} = 2^2 = 4$

Total pages = 5

Total size of P.T. across all levels = $5 * 4 \text{ KB}$
 $= 20 \text{ KB}$

ques)



Page size = $2^{11}B = 2KB$

no. of pages to store outer P.T. = 1

$$\begin{array}{r} \text{--- } 11 \text{ ---} \\ \text{Inner P.T.} = 2^4 = 16 \\ \hline \text{Total} = 17 \end{array}$$

$$\text{Total P.T. size} = 17 * 2KB = 34KB$$

Ques)

P_1	P_2	d
10	10	13

→ Page size = $2^{13} B = 8KB$

No. of pages to store outer P.T. = 1

— 11 ————— Inner P.T. = $2^{10} = 1024$

————— Total = 1025

$$\begin{aligned} \text{P.T. Size} &= 1025 * 8KB \\ &= \underline{\underline{8200 KB}} \end{aligned}$$



Topic : Question

Size of page tables across all levels?

P1	P2	P3	D
1	8	8	10

→ Page size = $2^{10} \text{ B} = 1 \text{ KB}$

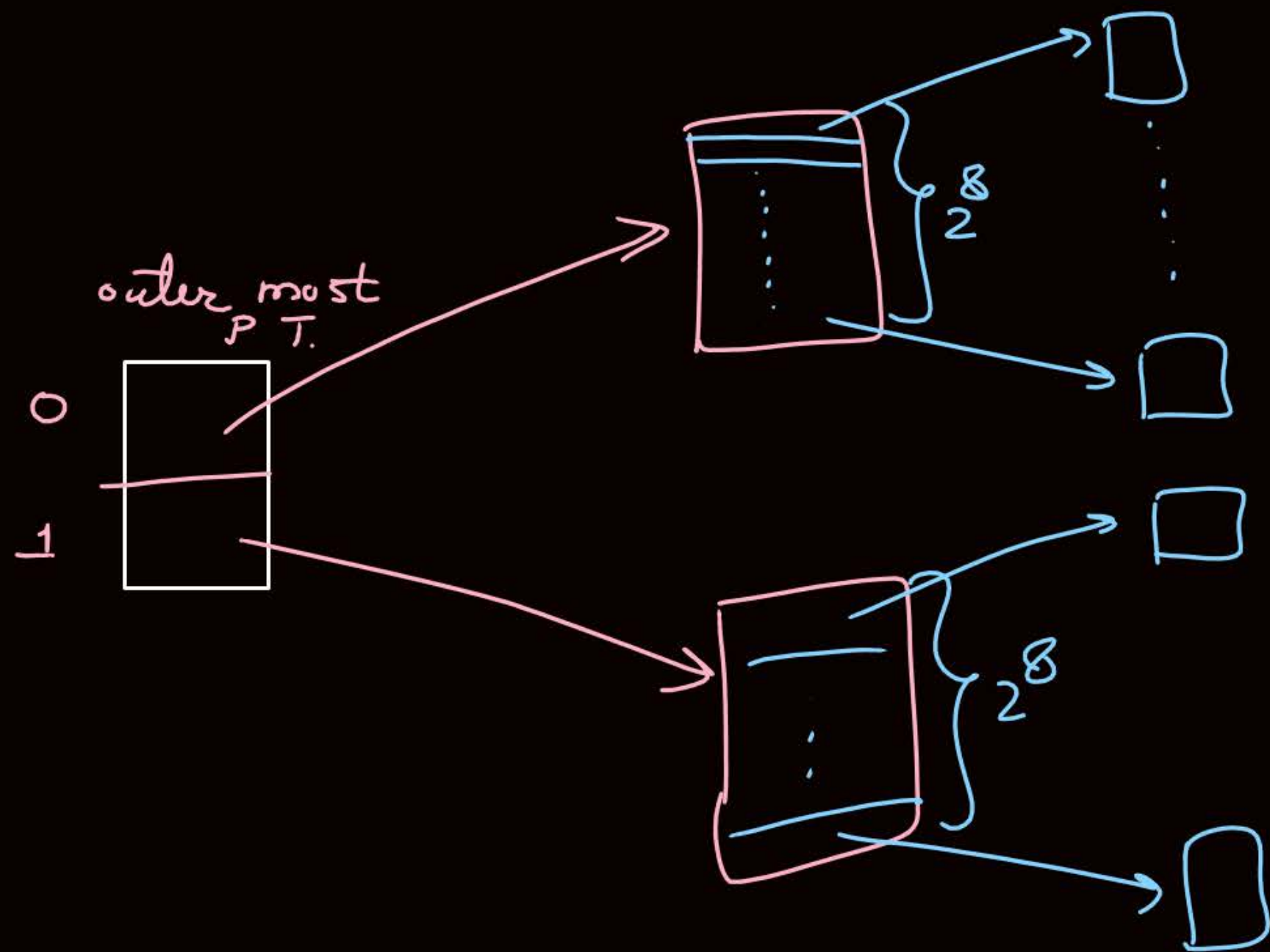
No. of pages to store outer most P.T. = 1

————||———— Middle P.T. = $2^{P_1} = 2^1 = 2$

————||———— Inner most P.T. = $2^{P_1} * 2^{P_2} = 2^1 * 2^8 = 512$

Total = 515

Total P.T. Size = $515 * 1 \text{ KB} = 515 \text{ KB}$



P_1	P_2	P_3	d
2	8	8	11

Page size = $2^{11} B = 2KB$

no. of pages in outermost P.T. = 1

—— 11 ——— middle — 11 — = $2^2 = 4$

—— 11 ——— innermost — 11 — = $2^2 * 2^8 = 1024$

Total = 1029

Total P.T. size = $1029 * 2KB$
 = 2058 KB

32-bits OS $\Rightarrow$ max V.A. = 32 bits

64-bits OS $\Rightarrow$ max V.A. = 64 bits



Topic : Question

H.W.

#Q. Size of page Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below:

←----- 39-bits Virtual Address ----->			
Level-1 Offset	Level-2 Offset	Level-3 Offset	Page Offset
9 bits	9 bits	9 bits	12 bits

The page size is 4 KB = (1KB = 2^{10} bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2 GB (1 GB = 2^{30} bytes) virtual memory which OS mapped to 2 GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB across all levels?



Topic : Question

H.W.

[GATE-2024]



#Q. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is _____?



Topic : Question

H.W.

[GATE-2024]



- #Q. A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is _____?





2 mins Summary



Topic

Multilevel Paging





Happy Learning

THANK - YOU

